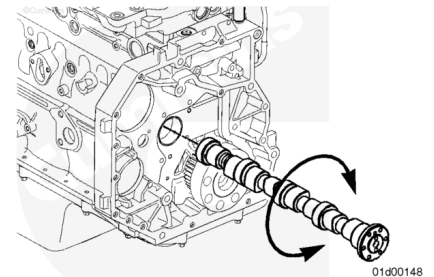


Trainings ▾

General Information

In order to remove the tappets from the engine, it is first necessary to start the removal process of the camshaft. Refer to the preparatory steps of the camshaft Procedure 001-008 (</qs3/pubsys2/xml/en/procedures/40/40-001-008-tr.html>), before following this procedure.

Note : Some of the graphics shown in this procedure are for rear gear train engines. However, the remove and install steps are the same for both rear gear train engines and front gear train engines.

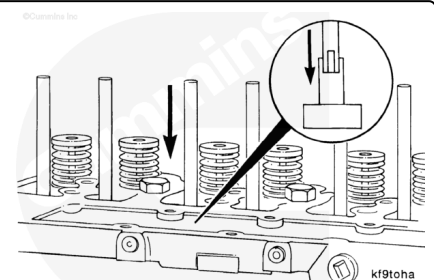


Remove

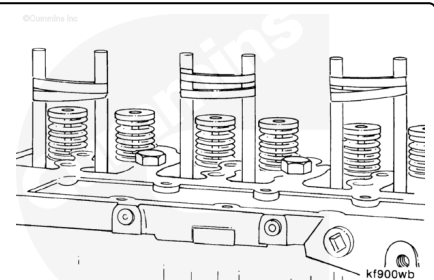
Use tappet removal tool kit, Part Number 3822513, to remove the tappets.

Push a wooden dowel rod into each tappet. It will probably be necessary to push the dowel into the tappet with a soft-face hammer.

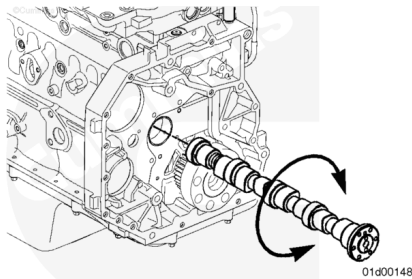
Note : When properly installed, the dowels can be used to pull the tappets up and should require considerable effort to be pulled out.



Pull each valve tappet up until it makes contact with the cylinder block. Put a rubber band around two dowels. This will hold the tappets up off the camshaft.

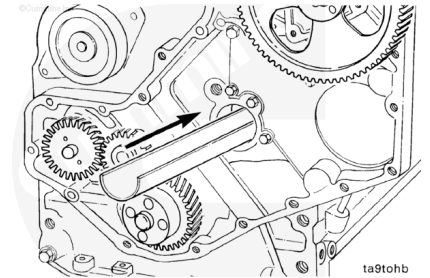


Remove the camshaft. Refer to Procedure 001-008 (</qs3/pubsys2/xml/en/procedures/40/40-001-008-tr.html>).



Insert the trough from the tappet replacement kit, Part Number 3822513, to the full length of the cam bore.

Note : Make sure the trough is positioned so that it will catch the tappet when the wooden dowel is removed.

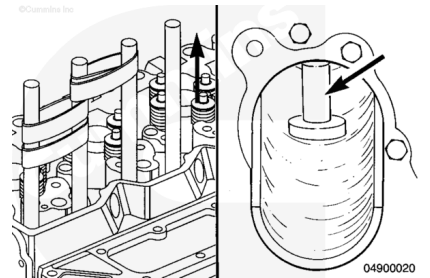


Only remove one tappet at a time.

Mark the position of the tappets as they are removed. The tappets **must** be installed in the same position when the engine is assembled.

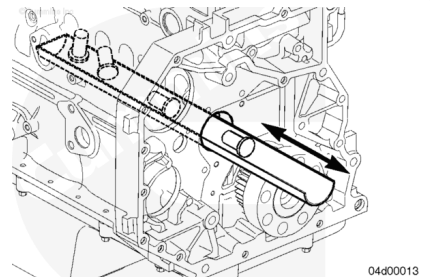
Remove the rubber band from the two companion tappets.

Secure the tappet that is **not** to be removed with the rubber band.

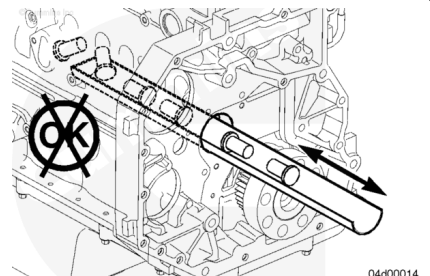


Note : When the tappet is dropped into the trough, most of the time it will fall over. However, if it does **not**, gently shake the trough just enough to allow the tappet to fall over before removing.

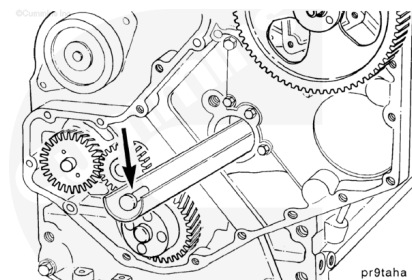
Pull the wooden dowel from the tappet bore, allowing the tappet to fall into the trough.



Take special care **not** to knock or shake the tappets over the end of the trough.



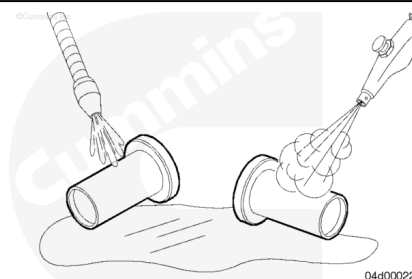
Carefully pull the trough and tappet from the cam bore, and remove the tappet. Repeat the process until all tappets are removed.



Clean and Inspect for Reuse

⚠ WARNING ⚠

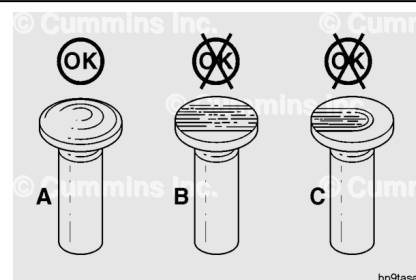
When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



Clean the tappets with solvent.

Inspect the socket, stem, and face for excessive wear, cracks, and other damage.

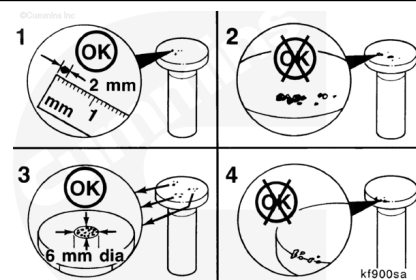
- A. Normal contact (exaggerated)
- B. Irregular contact - do **not** reuse
- C. Irregular contact - do **not** reuse.



Pit marks on the tappet face are acceptable.

The following criteria define the size of the pits allowed.

- A single pit can **not** be greater than 2 mm [0.079 in].
- Interconnection of pits is **not** allowed.
- Total pits when added together should **not** exceed 6-mm [0.236-in] diameter or a total of 4 percent of the tappet face.
- No pitting is allowed on the edges of the wear face of the tappet.

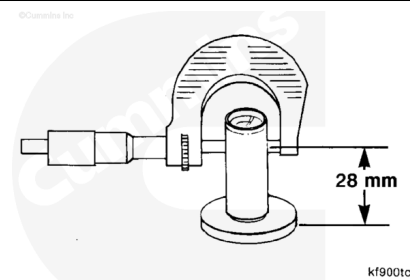


For detailed illustrations and reuse guidelines, refer to Service Bulletin 3666475

(/qs3/pubsys2/xml/en/bulletin/3666475.html), Camshaft and Tappet Reuse Guidelines.

Measure the valve tappet stem.

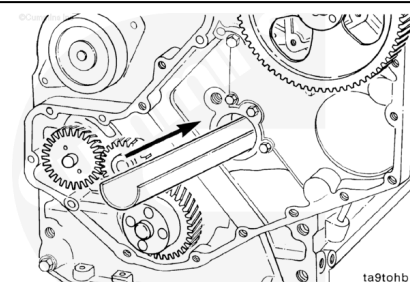
mm		in
15.936	MIN	0.627
15.977	MAX	0.629



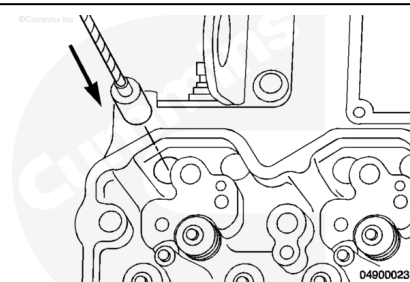
If the tappet stem is **not** within specifications, replace the tappet. Make sure to inspect the tappet bore for wear. Refer to Procedure 001-026 (</qs3/pubsys2/xml/en/procedures/40/40-001-026.html>).

Install

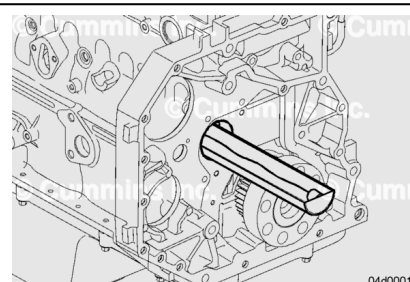
Insert the trough from the tappet replacement kit, Part Number 3822513, the full length of the cam bore.



Lower the tappet installation tool down through the push tube hole, through the tappet bore, and into the trough.



Feed the installation tool through the cam bores by carefully pulling the trough/installation tool out the front of the engine. The barrier at the rear of the trough will pull the tool out **most** of the time.

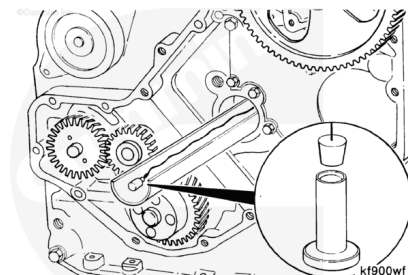


Lubricate the tappets with assembly lubricant, Part Number 3163087, or equivalent.

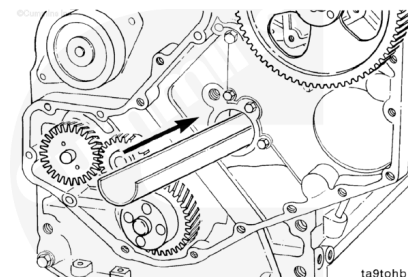


To aid in removing the installation tool after the tappet is installed, work the tool in and out of the tappet several times **before** installing the tappets.

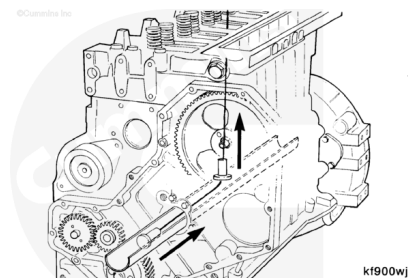
Insert the installation tool into the tappet.



Slide the trough into the cam bore.

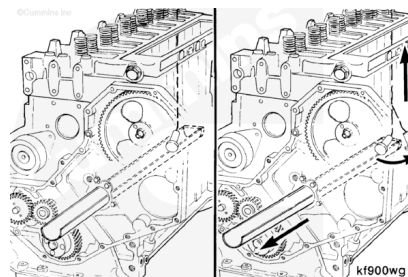


Pull the tool and tappet through the cam bore and up into the tappet bore.



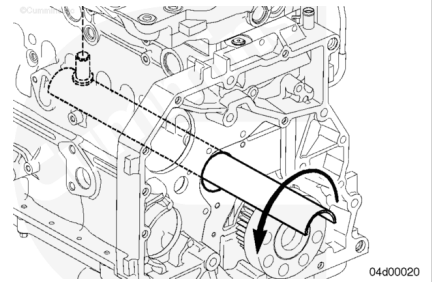
If difficulty is experienced in getting the tappet to make the bend from the trough up to the tappet bore, pull the trough out enough to allow the tappet to drop down and align itself.

Pull the tappet up into the bore.

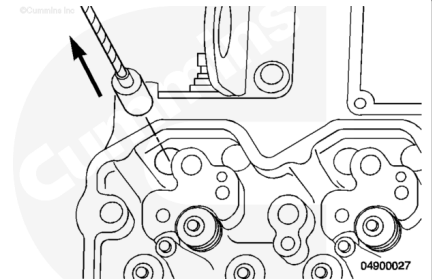


After the tappet has been pulled up into position, slide the trough back into the cam bore, and rotate it $\frac{1}{2}$ of a turn. This will position the round side of the trough up, which will hold the

tappet in place.



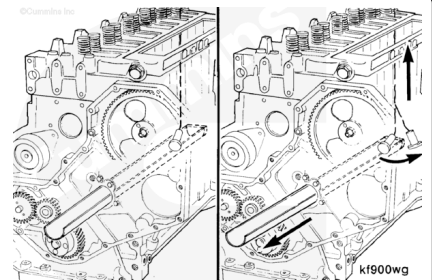
Remove the installation tool from the tappet.



Install a wooden dowel into the top of the tappet.

Wrap rubber bands around the wooden dowels to secure the tappets.

Repeat this process until all tappets have been installed.



Finishing Steps

- Install the camshaft. Refer to Procedure 001-008 (</qs3/pubsys2/xml/en/procedures/40/40-001-008-tr.html>).



